

HW2

Tables created:

```
fyousséf_db-> \dt
          List of relations
 Schema | Name   | Type  | Owner
-----+-----+-----+-----
 public | airline| table | fyousséf
 public | airport| table | fyousséf
 public | flight | table | fyousséf
 public | segment| table | fyousséf
(4 rows)
```

1.

Content within tables:

```
fyousséf_db=> \i hw2_1.sql
DROP TABLE
DROP TABLE
DROP TABLE
DROP TABLE
CREATE TABLE
CREATE TABLE
CREATE TABLE
CREATE TABLE
INSERT 0 6
INSERT 0 5
INSERT 0 6
INSERT 0 8
fyousséf_db=> SELECT * FROM airport;
 id | name | city | state | elevation
-----+-----+-----+-----+-----
 SEA | Seattle-Tacoma International | Seattle | WA | 433
 GEG | Spokane International | Spokane | WA | 2377
 LAX | Los Angeles International | Los Angeles | CA | 125
 DEN | Denver International | Denver | CO | 5434
 JFK | John F. Kennedy International | New York | NY | 13
 SFO | San Francisco International | San Francisco | CA | 13
(6 rows)
```

```
fyousséf_db=> SELECT * FROM airline;
 code | name | main_hub | yr_founded
-----+-----+-----+-----
 AS | Alaska Airlines | SEA | 1932
 UA | United Airlines | DEN | 1926
 WN | Southwest Airlines | LAX | 1967
 AA | American Airlines | JFK | 1930
 DL | Delta Air Lines | SFO | 1925
(5 rows)
```

```
fyoussef_db=> SELECT * FROM flight;
airline | flight_number | departure | arrival | flights_per_wk
-----+-----+-----+-----+-----
AS      |      212     | GEG      | SEA     |      21
AS      |      450     | SEA      | LAX     |      14
UA      |     1760     | DEN      | JFK     |      10
UA      |      567     | SFO      | DEN     |      18
WN      |      150     | SEA      | LAX     |       7
AA      |      320     | JFK      | SEA     |       6
(6 rows)
```

```
fyoussef_db=> SELECT * FROM segment;
airline | flight_number | segment_offset | start_airport | end_airport
-----+-----+-----+-----+-----
AS      |      212     |      1       | GEG           | SEA
AS      |      450     |      1       | SEA           | LAX
UA      |     1760     |      1       | DEN           | SFO
UA      |     1760     |      2       | SFO           | JFK
UA      |      567     |      1       | SFO           | DEN
WN      |      150     |      1       | SEA           | GEG
WN      |      150     |      2       | GEG           | LAX
AA      |      320     |      1       | JFK           | SEA
(8 rows)
```

Failing inserts:

airport:

```
psql -U fyoussef -h cps-postgresql.gonzaga.edu -d fyoussef_db -p 5432

fyoussef_db=> INSERT INTO airport VALUES ('SEA', 'Duplicate', 'Nowhere', 'WA', 100);
ERROR: duplicate key value violates unique constraint "airport_pkey"
DETAIL: Key (id)=(SEA) already exists.
fyoussef_db=> INSERT INTO airport VALUES ('PDX', NULL, 'Portland', 'OR', 30);
ERROR: null value in column "name" of relation "airport" violates not-null constraint
DETAIL: Failing row contains (PDX, null, Portland, OR, 30).
fyoussef_db=>
```

airline:

```
fyoussef_db=> INSERT INTO airline VALUES ('ZZ', 'Fake Air', 'XXX', 2000);
ERROR: insert or update on table "airline" violates foreign key constraint "airline_main_hub_fkey"
DETAIL: Key (main_hub)=(XXX) is not present in table "airport".
fyoussef_db=> INSERT INTO airline VALUES ('ZZ', 'Fake Air', 'XXX', 2000);
ERROR: insert or update on table "airline" violates foreign key constraint "airline_main_hub_fkey"
DETAIL: Key (main_hub)=(XXX) is not present in table "airport".
fyoussef_db=> |
```

Flight:

```
DETAIL: Key (main_hub) (XXX) is not present in table "airport".
fyoussef_db=> INSERT INTO flight VALUES ('AS',999,'SEA','SEA',5);
ERROR: new row for relation "flight" violates check constraint "flight_check"
DETAIL: Failing row contains (AS, 999, SEA, SEA, 5).
fyoussef_db=> INSERT INTO flight VALUES ('ZZ',1,'SEA','LAX',3);
ERROR: insert or update on table "flight" violates foreign key constraint "flight_airline_fkey"
DETAIL: Key (airline)=(ZZ) is not present in table "airline".
fyoussef_db=>
```

Segment:

```
DETAIL: Key (airline)=(ZZ) is not present in table "airline".
fyoussef_db=> INSERT INTO segment VALUES ('AS',450,0,'SEA','LAX');
ERROR: new row for relation "segment" violates check constraint "segment_segment_offset_check"
DETAIL: Failing row contains (AS, 450, 0, SEA, LAX).
fyoussef_db=> INSERT INTO segment VALUES ('AS',999,1,'SEA','LAX');
ERROR: insert or update on table "segment" violates foreign key constraint "segment_airline_flight_number_fkey"
DETAIL: Key (airline, flight_number)=(AS, 999) is not present in table "flight".
fyoussef_db=> |
```

2. Schema:

A. mgroup(name, formed_yr)

Primary key: name

Foreign key: N/A

B. musician(first_name, last_name, stage_name, birth_year)

Primary key: (first_name, last_name)

Foreign key: N/A

C. genre(label, description)

Primary key: label

Foreign key: N/A

D. label(name)

Primary key: name

Foreign key: N/A

E. album(group_name, title, rec_year, label_name)

Primary key: (group_name, title)

Foreign keys:

group_name → mgroup(name)

label_name → label(name)

F. song(title, written_y)

Primary key: title

Foreign key: N/A

G. group_genre(group_name, genre_label)

Primary key: (group_name, genre_label)

Foreign keys:

group_name → mgroup(name)

genre_label → genre(label)

H. influence(group_name, influenced_by)

Primary key: (group_name, influenced_by)

Foreign keys:

group_name → mgroup(name)

influenced_by → mgroup(name)

I. wrote(song_title, first_name, last_name)

Primary key: (song_title, first_name, last_name)

Foreign keys:

song_title → song(title)

(first_name, last_name) → musician(first_name, last_name)

J. track(track_id, song_title, rec_year)

Primary key: track_id

Foreign key:

song_title → song(title)

K. track_musician(track_id, first_name, last_name)

Primary key: (track_id, first_name, last_name)

Foreign keys:

track_id → track(track_id)

(first_name, last_name) → musician(first_name, last_name)

L. album_track(group_name, album_title, track_no, track_id)

Primary key: (group_name, album_title, track_no)

Foreign keys:

(group_name, album_title) → album(group_name, title)

track_id → track(track_id)

M. membership(group_name, first_name, last_name, start_year, end_year)

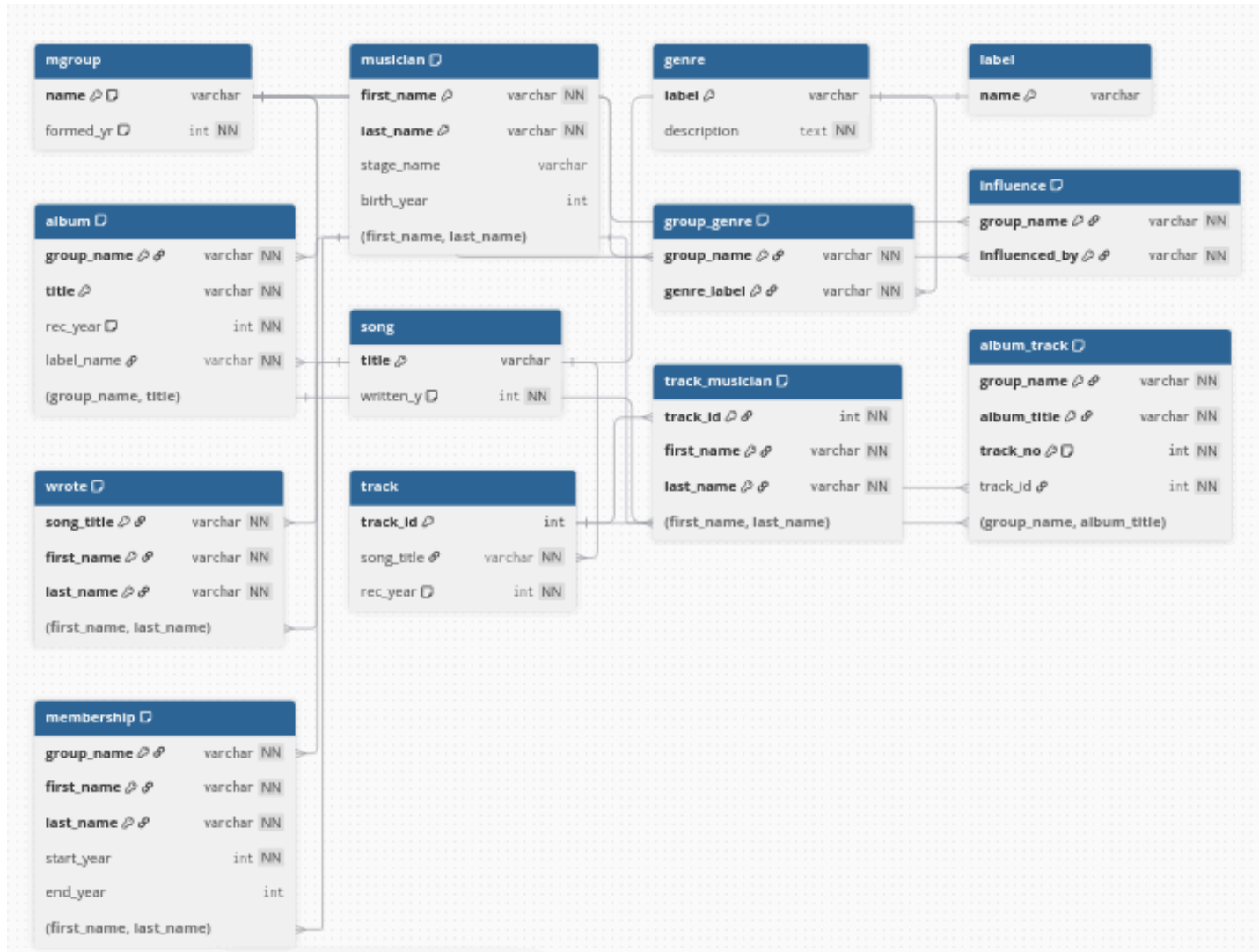
Primary key: (group_name, first_name, last_name)

Foreign keys:

group_name → mgroup(name)

(first_name, last_name) → musician(first_name, last_name)

Schema Diagram:



3. Tables created:

```
fyouussef_db=> \dt
               List of relations
 Schema |      Name      | Type  | Owner
-----+-----+-----+-----
 public | album           | table | fyouussef
 public | album_track     | table | fyouussef
 public | genre           | table | fyouussef
 public | group_genre     | table | fyouussef
 public | influence       | table | fyouussef
 public | label           | table | fyouussef
 public | membership      | table | fyouussef
 public | mgroup          | table | fyouussef
 public | musician        | table | fyouussef
 public | song            | table | fyouussef
 public | track           | table | fyouussef
 public | track_musician  | table | fyouussef
 public | wrote           | table | fyouussef
(13 rows)

fyouussef_db=>
```

Content within tables:

```
fyouussef_db=> SELECT * FROM album;
 group_name |      title      | rec_year | label_name
-----+-----+-----+-----
 The Beatles | Revolver        | 1966     | Parlophone
 Nirvana     | Nevermind       | 1991     | DGC
 Phoebe Bridgers | Punisher       | 2020     | Dead Oceans
 The Byrds   | Mr. Tambourine Man | 1965     | Columbia
 Boygenius   | boygenius (EP)  | 2018     | Interscope
(5 rows)

fyouussef_db=> SELECT * FROM album_track;
 group_name | album_title | track_no | track_id
-----+-----+-----+-----
 The Beatles | Revolver    | 2        | 1001
 Nirvana     | Nevermind   | 1        | 2001
 Phoebe Bridgers | Punisher   | 2        | 3001
 The Byrds   | Mr. Tambourine Man | 1        | 4001
 Boygenius   | boygenius (EP) | 3        | 5001
(5 rows)

fyouussef_db=> |
```

```
fyousssef_db=> SELECT * FROM genre;
  label | description
-----+-----
rock   | Guitar-driven popular music
folk rock | Folk + rock elements
indie   | Independent/alternative
pop     | Mainstream pop
alt     | Alternative
(5 rows)
```

```
fyousssef_db=> SELECT * FROM group_genre;
 group_name | genre_label
-----+-----
The Beatles | rock
Nirvana     | rock
The Byrds   | folk rock
Phoebe Bridgers | indie
Boygenius   | indie
(5 rows)
```

```
fyousssef_db=>
```

```
fyousssef_db=> SELECT * FROM influence;
 group_name | influenced_by
-----+-----
Nirvana     | The Beatles
The Beatles | The Byrds
(2 rows)
```

```
fyousssef_db=> SELECT * FROM label;
 name
-----
Parlophone
DGC
Columbia
Dead Oceans
Interscope
(5 rows)
```

```
fyousssef_db=> SELECT * FROM membership;
 group_name | first_name | last_name | start_year | end_year
-----+-----+-----+-----+-----
The Beatles | John      | Lennon   | 1960       | 1970
The Beatles | Paul      | McCartney | 1960       | 1970
Nirvana     | Kurt     | Cobain   | 1987       | 1994
Nirvana     | Dave     | Grohl    | 1990       | 1994
Phoebe Bridgers | Phoebe   | Bridgers | 2014       |
(5 rows)
```

```
fyousssef_db=>
```

```

fyousssef_db=> SELECT * FROM mgroup;
      name      | formed_yr
-----+-----
The Beatles    | 1960
Nirvana        | 1987
The Byrds      | 1964
Phoebe Bridgers | 2014
Boygenius      | 2018
(5 rows)

fyousssef_db=> SELECT * FROM musician;
first_name | last_name | stage_name | birth_year
-----+-----+-----+-----
John       | Lennon   |            | 1940
Paul       | McCartney |           | 1942
Kurt       | Cobain   |           | 1967
Dave       | Grohl    |           | 1969
Phoebe     | Bridgers |           | 1994
(5 rows)

fyousssef_db=> SELECT * FROM song;
      title      | written_y
-----+-----
Eleanor Rigby   | 1966
Smells Like Teen Spirit | 1991
Kyoto           | 2020
Mr. Tambourine Man | 1964
Me & My Dog     | 2018
(5 rows)

fyousssef_db=>

```

```

fyousssef_db=> SELECT * FROM track;
track_id | song_title | rec_year
-----+-----+-----
1001     | Eleanor Rigby | 1966
2001     | Smells Like Teen Spirit | 1991
3001     | Kyoto | 2020
4001     | Mr. Tambourine Man | 1965
5001     | Me & My Dog | 2018
(5 rows)

fyousssef_db=> SELECT * FROM track_musician;
track_id | first_name | last_name
-----+-----+-----
1001     | John      | Lennon
1001     | Paul      | McCartney
2001     | Kurt      | Cobain
2001     | Dave      | Grohl
3001     | Phoebe    | Bridgers
(5 rows)

fyousssef_db=> SELECT * FROM wrote;
      song_title      | first_name | last_name
-----+-----+-----
Eleanor Rigby        | Paul      | McCartney
Smells Like Teen Spirit | Kurt      | Cobain
Kyoto                 | Phoebe    | Bridgers
Mr. Tambourine Man    | John      | Lennon
(4 rows)

fyousssef_db=>

```


Failing inserts:

```
psql -U fyoussef -h cps-postgresql.gonzaga.edu -d fyoussef_db — psql -U fyoussef -h cps-postgresql.gonzaga.edu -d fyoussef_db

INSERT 0 5
psql:hw2.3.sql:228: ERROR:  duplicate key value violates unique constraint "mgroun_pkey"
DETAIL:  Key (name)=(Nirvana) already exists.
psql:hw2.3.sql:230: ERROR:  null value in column "name" of relation "mgroun" violates not-null constraint
DETAIL:  Failing row contains (null, 2000).
psql:hw2.3.sql:233: ERROR:  duplicate key value violates unique constraint "musician_pkey"
DETAIL:  Key (first_name, last_name)=(John, Lennon) already exists.
psql:hw2.3.sql:234: ERROR:  null value in column "first_name" of relation "musician" violates not-null constraint
DETAIL:  Failing row contains (null, Lennon, null, 1940).
psql:hw2.3.sql:238: ERROR:  duplicate key value violates unique constraint "genre_pkey"
DETAIL:  Key (label)=(rock) already exists.
psql:hw2.3.sql:240: ERROR:  null value in column "description" of relation "genre" violates not-null constraint
DETAIL:  Failing row contains (noise, null).
psql:hw2.3.sql:244: ERROR:  duplicate key value violates unique constraint "label_pkey"
DETAIL:  Key (name)=(Parlophone) already exists.
psql:hw2.3.sql:246: ERROR:  null value in column "name" of relation "label" violates not-null constraint
DETAIL:  Failing row contains (null).
psql:hw2.3.sql:250: ERROR:  duplicate key value violates unique constraint "album_pkey"
DETAIL:  Key (group_name, title)=(The Beatles, Revolver) already exists.
psql:hw2.3.sql:252: ERROR:  insert or update on table "album" violates foreign key constraint "album_label_name_fkey"
DETAIL:  Key (label_name)=(NotARealLabel) is not present in table "label".
psql:hw2.3.sql:256: ERROR:  duplicate key value violates unique constraint "song_pkey"
DETAIL:  Key (title)=(Kyoto) already exists.
psql:hw2.3.sql:258: ERROR:  null value in column "title" of relation "song" violates not-null constraint
DETAIL:  Failing row contains (null, null).
psql:hw2.3.sql:262: ERROR:  duplicate key value violates unique constraint "group_genre_pkey"
DETAIL:  Key (group_name, genre_label)=(The Beatles, rock) already exists.
psql:hw2.3.sql:264: ERROR:  insert or update on table "group_genre" violates foreign key constraint "group_genre_group_name_fkey"
```

```
DETAIL:  Key (group_name)=(NoGroup) is not present in table "mgroun".
psql:hw2.3.sql:268: ERROR:  new row for relation "influence" violates check constraint "influence_check"
DETAIL:  Failing row contains (Nirvana, Nirvana).
psql:hw2.3.sql:270: ERROR:  insert or update on table "influence" violates foreign key constraint "influence_group_name_fkey"
DETAIL:  Key (group_name)=(NoGroup) is not present in table "mgroun".
psql:hw2.3.sql:274: ERROR:  insert or update on table "wrote" violates foreign key constraint "wrote_first_name_last_name_fkey"
DETAIL:  Key (first_name, last_name)=(Taylor, Swift) is not present in table "musician".
psql:hw2.3.sql:276: ERROR:  duplicate key value violates unique constraint "wrote_pkey"
DETAIL:  Key (song_title, first_name, last_name)=(Eleanor Rigby, Paul, McCartney) already exists.
psql:hw2.3.sql:280: ERROR:  duplicate key value violates unique constraint "track_pkey"
DETAIL:  Key (track_id)=(1001) already exists.
psql:hw2.3.sql:282: ERROR:  insert or update on table "track" violates foreign key constraint "track_song_title_fkey"
DETAIL:  Key (song_title)=(Nonexistent Song) is not present in table "song".
psql:hw2.3.sql:286: ERROR:  duplicate key value violates unique constraint "track_musician_pkey"
DETAIL:  Key (track_id, first_name, last_name)=(1001, John, Lennon) already exists.
psql:hw2.3.sql:288: ERROR:  insert or update on table "track_musician" violates foreign key constraint "track_musician_track_id_fkey"
DETAIL:  Key (track_id)=(9999) is not present in table "track".
psql:hw2.3.sql:292: ERROR:  duplicate key value violates unique constraint "album_track_pkey"
DETAIL:  Key (group_name, album_title, track_no)=(The Beatles, Revolver, 2) already exists.
psql:hw2.3.sql:294: ERROR:  insert or update on table "album_track" violates foreign key constraint "album_track_group_name_album_title_fkey"
DETAIL:  Key (group_name, album_title)=(The Beatles, Help!) is not present in table "album".
psql:hw2.3.sql:298: ERROR:  new row for relation "membership" violates check constraint "membership_check"
DETAIL:  Failing row contains (Nirvana, Dave, Grohl, 1994, 1993).
psql:hw2.3.sql:300: ERROR:  insert or update on table "membership" violates foreign key constraint "membership_group_name_fkey"
DETAIL:  Key (group_name)=(Unknown) is not present in table "mgroun".
fyoussef_db=>
```